Journal Pre-proof



Comparative Outcomes of Endovascular and Open Surgical Repair for Extracranial Carotid Artery Aneurysms: A Systematic Review and Meta-analysis

Mayara Leite Coutinho, MD, Celso Ricardo Bregalda Neves, MD, pHD, Inez Ohashi Torres, MD, pHD, Lucas Monteiro Delgado, Mario D'Oria, MD, Antonio Eduardo Zerati, MD, PhD, Pedro Puech Leão, MD, pHD

PII: S0890-5096(26)00729-6

DOI: https://doi.org/10.1016/j.avsg.2026.09.065

Reference: AVSG 8633

To appear in: *Annals of Vascular Surgery*

Received Date: 29 August 2026

Revised Date: 29 September 2026

Accepted Date: 29 September 2026

Please cite this article as: Coutinho ML, Neves CRB, Torres IO, Delgado LM, D'Oria M, Zerati AE, Leão PP, Comparative Outcomes of Endovascular and Open Surgical Repair for Extracranial Carotid Artery Aneurysms: A Systematic Review and Meta-analysis, *Annals of Vascular Surgery* (2026), doi: https://doi.org/10.1016/j.avsg.2026.09.065.

This is a PDF of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability. This version will undergo additional copyediting, typesetting and review before it is published in its final form. As such, this version is no longer the Accepted Manuscript, but it is not yet the definitive Version of Record; we are providing this early version to give early visibility of the article. Please note that Elsevier's sharing policy for the Published Journal Article applies to this version, see: https://www.elsevier.com/about/policies-and-standards/sharing#4-published-journal-article. Please also note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2026 The Author(s). Published by Elsevier Inc.

Journal Pre-proof

Comparative Outcomes of Endovascular and Open Surgical Repair for Extracranial Carotid Artery Aneurysms: A Systematic Review and Meta-analysis

Running Short Title

Endovascular vs Open Repair of ECAAs

Authors

Mayara Leite Coutinho MD[1], Celso Ricardo Bregalda Neves MD, pHD[1], Inez Ohashi Torres MD, pHD[1], Lucas Monteiro Delgado[2], Mario D'Oria MD[3], Antonio Eduardo Zerati MD, PhD[1] , Pedro Puech Leão MD, pHD[1]

Affiliations

[1]Department of Vascular Surgery, Hospital das Clínicas HCFMUSP, Faculdade de Medicina, Universidade de São Paulo, São Paulo, Brazil

[2] Universidade Federal de Minas Gerais, Minas Gerais, Brazil.

[3]Division of Vascular and Endovascular Surgery, Department of Clinical Surgical and Health Sciences, University of Trieste, Trieste, Italy.

Corresponding author

Mayara Leite Coutinho

Address: Street , 255 - Dr. Enéas Carvalho de Aguiar, São Paulo - SP, 05405-000

Phone number: (+55) 11 933307000

E-mail address: mayara.leite84@hotmail.com

Funding and Competing Interests Statement

This study received no financial or non-financial support, and the authors declare no conflicts of interest. No datasets were generated during the conduct of this review. Additional data and materials related to the analysis are available upon reasonable request. This includes the standardized data extraction form, the data extracted from the included studies, and the analytic code (R, version 4.5) used to perform the meta-analyses, all of which are available from the corresponding author upon reasonable request.

Journal Pre-proof

CRediT Authorship Contribution Statement

Mayara Leite Coutinho: Conceptualization, Methodology, Data curation, Formal analysis, Investigation, Software, Visualization, Project administration, Writing – original draft.

Celso Ricardo Bregalda Neves: Conceptualization, Methodology, Data curation, Investigation, Validation, Supervision, Writing – review & editing.

Inez Ohashi Torres: Conceptualization, Methodology, Data curation, Investigation, Validation, Supervision, Writing – review & editing.

Lucas Monteiro Delgado: Investigation, Data curation, Validation, Writing – review & editing.

Mario D'Oria: Methodology, Formal analysis, Validation, Writing – review & editing.

Antonio Eduardo Zerati: Methodology, Validation, Supervision, Writing – review & editing.

Pedro Puech Leão: Supervision, Writing – review & editing.

Journal Pre-proof

What this study adds

This systematic review and meta-analysis provides an updated quantitative comparison of endovascular and open surgical repair for extracranial carotid artery aneurysms. Both approaches were associated with low perioperative mortality and stroke rates, low reintervention rates, and high primary

patency, with no significant differences identified in comparative analyses. Although endovascular repair appears to be a safe alternative in appropriately selected patients, open repair is supported by a larger evidence base and longer follow-up. Importantly, the available evidence is observational and subject to substantial confounding by indication; therefore, comparable observed outcomes should not be interpreted as therapeutic equivalence.

Journal Pre-proof

Comparative Outcomes of Endovascular and Open Surgical Repair for Extracranial Carotid Artery Aneurysms: A Systematic Review and Meta-analysis

Running Short Title

Endovascular vs Open Repair of ECAAs

Authors

Mayara Leite Coutinho MD[1], Celso Ricardo Bregalda Neves MD, pHD[1], Inez Ohashi Torres MD, pHD[1], Lucas Monteiro Delgado[2], Mario D'Oria MD[3], Antonio Eduardo Zerati MD, PhD[1] , Pedro Puech Leão MD, pHD[1]

Affiliations

[1]Department of Vascular Surgery, Hospital das Clínicas HCFMUSP, Faculdade de Medicina, Universidade de São Paulo, São Paulo, Brazil

[2] Universidade Federal de Minas Gerais, Minas Gerais, Brazil.

[3]Division of Vascular and Endovascular Surgery, Department of Clinical Surgical and Health Sciences, University of Trieste, Trieste, Italy.

Corresponding author

Mayara Leite Coutinho

Address: Street , 255 - Dr. Enéas Carvalho de Aguiar, São Paulo - SP, 05405-000

Phone number: (+55) 11 933307000

E-mail address: mayara.leite84@hotmail.com

Funding and Competing Interests Statement

This study received no financial or non-financial support, and the authors declare no conflicts of interest. No datasets were generated during the conduct of this review. Additional data and materials related to the analysis are available upon reasonable request. This includes the standardized data extraction form, the

Journal Pre-proof

data extracted from the included studies, and the analytic code (R, version 4.5) used to perform the meta-analyses, all of which are available from the corresponding author upon reasonable request.

CRediT Authorship Contribution Statement

Mayara Leite Coutinho: Conceptualization, Methodology, Data curation, Formal analysis, Investigation, Software, Visualization, Project administration, Writing – original draft.

Celso Ricardo Bregalda Neves: Conceptualization, Methodology, Data curation, Investigation, Validation, Supervision, Writing – review & editing.

Inez Ohashi Torres: Conceptualization, Methodology, Data curation, Investigation, Validation, Supervision, Writing – review & editing.

Lucas Monteiro Delgado: Investigation, Data curation, Validation, Writing – review & editing.

Mario D'Oria: Methodology, Formal analysis, Validation, Writing – review & editing.

Antonio Eduardo Zerati: Methodology, Validation, Supervision, Writing – review & editing.

Pedro Puech Leão: Supervision, Writing – review & editing.

Journal Pre-proof

ARTICLE HIGHLIGHTS

Type of Research: Systematic review and meta-analysis of comparative and single-arm observational studies.

Key Findings: Among 26 retrospective studies (707 patients, 725 procedures), endovascular repair (ER) and open surgical repair (OpR) of extracranial carotid artery aneurysms showed no significant differences in 30-day mortality (OR, 2.58; 95% CI, 0.53–12.52) or 30-day stroke (OR, 1.08; 95% CI, 0.30–3.91). Pooled single-arm 30-day mortality and stroke rates were 1.18% and 4.29% after OpR, versus 0% and 1.25% after ER; primary patency was 98.44% after OpR and 94.50% after ER.

Take Home Message: Endovascular and open repair of extracranial carotid artery aneurysms showed similar perioperative event rates and similar observed late event rates in observational data, but heterogeneous follow-up and the absence of time-to-event analyses preclude a formal comparison of long-term durability. Open repair remains supported by a larger and longer-followed evidence base, while endovascular repair appears to be a feasible alternative in appropriately selected patients, although evidence on its long-term durability remains limited.

TABLE OF CONTENTS SUMMARY

This systematic review and meta-analysis of 26 studies (707 patients) found similar perioperative outcomes and similar observed late event rates between endovascular and open surgical repair of extracranial carotid artery aneurysms, although heterogeneous follow-up, the absence of time-to-event analyses, and the predominantly moderate to serious risk of bias of the available evidence limit definitive conclusions about long-term durability.

Journal Pre-proof

ABSTRACT

Background: Extracranial carotid artery aneurysms (ECAAs) are rare lesions managed by open surgical repair (OpR) or endovascular repair (ER). We compared the perioperative and long-term outcomes of both strategies.

Methods: A PRISMA-compliant systematic review and meta-analysis, registered in PROSPERO, was performed. Comparative studies were pooled using random-effects models, and separate single-arm meta-analyses were conducted for OpR and ER.

Results: Twenty-six retrospective studies involving 707 patients and 725 procedures were included (574 OpR and 151 ER). Six comparative studies (228 procedures) showed no significant differences between ER and OpR in 30-day mortality (OR, 2.58; 95% CI, 0.53–12.52), 30-day stroke (OR, 1.08; 95% CI, 0.30–3.91), late stroke, reintervention, or primary patency. In the single-arm analyses, pooled 30-day mortality and stroke rates after OpR were 1.18% and 4.29%, respectively. No 30-day deaths and one stroke were reported after ER, yielding a pooled stroke rate of 1.25% (95% CI, 0.18–8.34). Primary patency was high after both OpR (98.44%) and ER (94.50%).

Conclusions: Observational evidence suggests similar perioperative outcomes and similar observed late event rates for ER and OpR in ECAAs, although heterogeneous follow-up and the absence of time-to-event analyses limit formal comparison of long-term durability. Both approaches were associated with low mortality, stroke, reintervention rates, and excellent primary patency. Open repair remains supported by a larger body of evidence and longer follow-up, whereas evidence for ER remains comparatively limited. Because treatment allocation in the available studies was influenced by patient characteristics, anatomical complexity, and surgical risk, the observed similarity in outcomes should not be interpreted as evidence of

therapeutic equivalence. Given the observational design and the predominantly moderate to serious risk of bias of the included studies, these findings should be interpreted with caution.

Registration: PROSPERO CRD420261433965.

Keywords: extracranial carotid artery aneurysm; endovascular repair; open surgical repair; meta-analysis; systematic review.

ABBREVIATIONS

CI: Confidence Interval

PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses

ROBINS-I: Risk Of Bias In Non-randomised Studies - of Interventions

OR: Odds Ratio

OpR: Open Repair

ER: Endovascular repair

ESVS: European Society for Vascular Surgery

ECAAs: Extracranial carotid artery aneurysms

Journal Pre-proof

INTRODUCTION

Extracranial carotid artery aneurysms (ECAAs) represent a rare vascular pathology, accounting for approximately 0.4% to 4% of all extracranial arterial aneurysms[1]. Despite their rarity, ECAAs are associated with significant morbidity and mortality when left untreated, with stroke being the most feared complication. The clinical importance of these lesions necessitates timely therapeutic intervention in most cases.

The clinical presentation of ECAAs is heterogeneous. Approximately 45% of patients present with a pulsatile neck mass, 26% manifest neurological deficits including transient ischemic attacks or stroke, while 23% remain asymptomatic at diagnosis[2]. ECAAs can be classified as true aneurysms (52% of cases) or pseudoaneurysms (45%), with diverse etiologies including atherosclerosis (34.2%), trauma (10%), vasculitis (5.57%), fibromuscular dysplasia, and connective tissue disorders[1,2]. The Attigah anatomical classification system identifies five types based on location, with Type I (43%) and Type III (28%) being most prevalent, which directly influences treatment selection[2].

Historically, open surgical repair has been considered the gold standard treatment for ECAAs, employing techniques such as resection with primary anastomosis, interposition grafting, or bypass procedures. However, with technological advances in endovascular therapy over the past two decades, minimally invasive techniques—including covered stents, overlapping closed-cell stents, flow-diverting braided stents, and stent-assisted coil embolization—have emerged as alternative therapeutic options[3].

Current evidence suggests that both approaches yield acceptable outcomes in terms of early mortality and cerebrovascular events. Comparative studies demonstrate 30-day mortality rates of 4% for open repair versus 0% for endovascular treatment, and stroke/transient ischemic attack rates of 5.5% versus 1.2%, respectively, although these differences have not reached statistical significance[4]. However, the complication profiles differ substantially: open repair is associated with higher rates of cranial nerve

injury (9-25%) compared to endovascular treatment (0%), while endovascular approaches may carry greater risk of reintervention due to restenosis or stent migration[2,4–6].

Despite accumulating evidence, the existing literature consists predominantly of retrospective case series and systematic reviews of small cohorts, with significant heterogeneity in surgical techniques, endovascular devices, patient selection criteria, and follow-up duration. No established consensus exists regarding which therapeutic approach offers superior outcomes for different patient subgroups when considering factors such as anatomical location, aneurysm type (true versus pseudoaneurysm), etiology, and clinical presentation. Furthermore, recent data from the International Carotid Aneurysm Registry have revealed concerning peri-procedural morbidity and mortality rates with endovascular treatment (7% stroke rate, 2% mortality within 30 days), despite excellent long-term outcomes including 100% stent patency and 98% aneurysm exclusion at median 65-month follow-up[6].

Therefore, a comprehensive meta-analysis comparing endovascular versus open repair in patients with extracranial carotid aneurysms is warranted to synthesize available evidence, quantify differences in clinical outcomes, identify patient subgroups who may preferentially benefit from a specific approach, and provide evidence-based guidance for clinical decision-making in this rare but potentially devastating condition.

MATERIALS AND METHODS

Protocol and registration

We performed the systematic review and meta-analysis according to the Cochrane Handbook for Systematic Reviews of Interventions and structured it according to the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines[7,8]. The study protocol was registered in the International Prospective Register of Systematic Reviews (PROSPERO) under registration number CRD420261433965[9]. The full protocol is publicly accessible at https://www.crd.york.ac.uk/prospero/display_record.php?ID=CRD420261433965. No amendments were made to the eligibility criteria, outcomes, or statistical methods after registration; the review was conducted as specified in the registered protocol.

Journal Pre-proof

Eligibility criteria

Inclusion in this meta-analysis was limited to studies that met all of the following eligibility criteria: (1) enrolled adults (≥18 years old) diagnosed with ECAA; (2) evaluated open surgical repair, endovascular repair, or directly compared both treatment modalities; and (3) reported at least one outcome of interest, including perioperative mortality, stroke, cranial nerve injury, reintervention, patency, or other clinically relevant outcomes.

Studies were excluded if they met any of the following criteria: (1) were published before 1996, as they predated contemporary carotid reconstruction techniques and standardized outcome reporting; (2) included intracranial carotid artery aneurysms, which represent a distinct disease entity with different pathophysiology, clinical presentation, and management strategies; (3) included carotid pseudoaneurysms exclusively secondary to trauma, infection, or iatrogenic injury without allowing separate extraction of data for true extracranial carotid artery aneurysms; (4) included mixed populations of extracranial carotid artery aneurysms and other supra-aortic trunk aneurysms without separate reporting of outcomes for carotid lesions; (5) did not provide extractable outcome data according to treatment modality; (6) lacked full-text availability; or (7) were case reports, small case series (<5 patients), conference abstracts without sufficient data, trial registrations without available results, reviews, meta-analyses, editorials, letters, expert opinions, or animal studies.

Search strategy and study selection

We systematically searched PubMed, Embase, and Cochrane Library databases from inception to June 21, 2026. The review was conducted according to the following PICO framework: **P** (Population), adults with extracranial carotid artery aneurysms; **I** (Intervention), endovascular repair; **C** (Comparator), open surgical repair; and **O** (Outcomes), perioperative mortality, stroke, cranial nerve injury, reintervention,

primary patency, and other clinically relevant postoperative outcomes. The complete search strategies for all databases are provided in **Supplementary Table I.**

We also searched the references of the included studies and previous systematic reviews and meta-analyses aiming for the inclusion of additional studies[10]. Two authors (M.L.C and C.R.B.N) independently conducted the search, imported results into Rayyan, a web-based systematic review tool, and triaged the studies. After the exclusion of duplicates and titles/abstracts unrelated to the clinical question, the eligibility of each remaining study was assessed based on the review of the full-text articles. Disagreements were solved by consensus

Data extraction

Two authors (M.L.C. and C.R.B.N) independently extracted data from the included studies using a standardized data extraction form. The following information was collected: (1) **study characteristics**, including first author, year of publication, country, study design, study period, sample size, number of patients undergoing open and endovascular repair, and follow-up duration; (2) **baseline demographic and aneurysm characteristics**, including mean age, sex, cardiovascular risk factors (hypertension, diabetes mellitus, dyslipidemia, coronary artery disease, smoking status, and previous stroke or transient ischemic attack), aneurysm type (true aneurysm or pseudoaneurysm), symptom status, mean aneurysm diameter, anatomical location (common, internal, or external carotid artery), bilateral involvement, and presence of intraluminal thrombus; (3) **treatment characteristics**, including the type of open surgical reconstruction (aneurysmectomy with end-to-end anastomosis, interposition graft, vein graft, prosthetic graft, bypass reconstruction, or carotid ligation) and endovascular repair (covered stent, bare-metal stent, flow diverter, or coil embolization), as well as technical success rates; and (4) **clinical outcomes of interest**, including 30 day mortality, 30 day stroke, late stroke, reintervention and patency, as predefined in the study protocol.

Two studies (Xu et al. and Obayi et al.) included patients who underwent hybrid repair, combining endovascular and open surgical techniques. For these studies, the denominator used in each pooled analysis was selected according to outcome type: patient-level outcomes (30-day mortality, 30-day stroke) were calculated using only patients who underwent pure endovascular treatment, whereas lesion-level

outcomes (technical success) could include all treated aneurysms. Patients who underwent hybrid repair were excluded from the single-arm endovascular outcome analyses and are presented descriptively in **Tables I and II** only.

Outcomes of interest

The primary outcomes of interest were 30-day mortality and 30-day stroke. Secondary outcomes included late stroke, reintervention, patency.

Statistical analysis

For the primary comparative meta-analysis, studies directly comparing endovascular and open surgical repair were pooled using odds ratios (ORs) with 95% confidence intervals (CIs) for binary outcomes and mean differences (MDs) with 95% CIs for continuous outcomes. Random-effects models were applied using the Mantel-Haenszel method for binary comparative outcomes, with between-study variance (τ^2) estimated by restricted maximum likelihood (REML) rather than the DerSimonian-Laird estimator[11,12]. Because several comparative outcomes included studies with zero events in one or both treatment arms, a continuity correction of 0.5 was applied to all studies within the affected analyses. Double-zero studies were retained in the primary analysis through the allstudies = TRUE specification (R package meta). Given the potential influence of this approach in sparse-event datasets, an alternative Peto odds ratio analysis without continuity correction was performed as a sensitivity analysis for outcomes affected by double-zero studies (see Sensitivity analysis and Results). The descriptive tables summarize all patients and procedures included in the systematic review. However, for each outcome, only studies and patients with sufficient outcome specific data were included in the corresponding meta-analysis. Consequently, the number of patients and procedures contributing to each pooled analysis may differ from the totals reported in the descriptive tables.

In addition, secondary single-arm meta-analyses were conducted separately for studies reporting outcomes after endovascular repair and for studies reporting outcomes after open surgical repair. For these

analyses, pooled event rates and corresponding 95% CIs were calculated using generalized linear mixed models with logit transformation.

Between-study heterogeneity was assessed using the Cochran Q test and the I^2 statistic, with heterogeneity considered significant when $P < 0.10$ and $I^2 > 25\%$[11,13]. When significant heterogeneity was detected, leave-one-out sensitivity analyses were performed by sequentially excluding individual studies to evaluate the robustness of the pooled estimates and determine whether the results were disproportionately influenced by any single study.

The predefined clinical outcomes included perioperative mortality, stroke, transient ischemic attack (TIA), cranial nerve injury, myocardial infarction, bleeding complications, wound complications, access-site complications, reintervention, graft or stent occlusion, restenosis, aneurysm recurrence, primary patency, secondary patency, late stroke, and late mortality.

All statistical analyses were performed using R statistical software (version 4.5; R Foundation for Statistical Computing, Vienna, Austria).

Sensitivity analysis

A leave-one-out sensitivity analysis was performed to assess the robustness of the pooled estimates. For the primary comparative meta-analysis, each study was sequentially excluded and the meta-analysis was repeated for all outcomes comparing endovascular and open surgical repair. This approach was used to evaluate the influence of individual studies on the overall effect estimates and to determine whether the pooled results were driven by any single study.

For the secondary single-arm meta-analyses, leave-one-out sensitivity analyses were performed for 30-day mortality and 30-day stroke separately in the endovascular repair cohort and the open surgical repair cohort. The pooled event rates were recalculated after sequential removal of each study to assess the stability of the findings.

Journal Pre-proof

Risk of bias assessment

Two independent reviewers (M.L.C. and L.M.D.) assessed the Risk of Bias In Non Randomized Studies of Interventions (ROBINS-I V2) tool[14]. Disagreements were resolved through consensus. No formal assessment of publication bias was performed because fewer than 10 comparative studies were available for the outcomes of interest.

Certainty assessment

The certainty of the evidence for the two primary outcomes (30-day mortality and 30-day stroke) from the primary comparative meta-analysis was assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach, considering risk of bias, inconsistency, indirectness, imprecision, and publication bias. Imprecision was judged primarily by the width of the 95% confidence interval relative to thresholds of appreciable benefit or harm and by the total number of events, rather than by a formal optimal information size calculation. Two reviewers (M.L.C. and C.R.B.N.) independently rated the certainty of evidence for each outcome, with disagreements resolved by consensus. Results are presented as a GRADE Summary of Findings table in **Supplementary Table II.**

RESULTS

Study selection

As detailed in **Figure 1**, the initial search identified 1,224 records. After the removal of 254 duplicate records and screening of titles and abstracts, 76 studies were selected for full-text review according to the prespecified eligibility criteria. Of these, 50 studies were excluded after full-text assessment for the following reasons: conference abstract only (n = 26), publication before 1996 (n = 7), non-English publication (n = 6), overlap with a more comprehensive cohort or duplicate population (n = 5), full text not available (n = 2), outcomes not stratified by treatment modality (n = 1), insufficient baseline data for quantitative synthesis (n = 1), outcomes not relevant to the review question (n = 1), and other reasons (editorial) (n = 1). A total of 26 retrospective studies were included in the final analysis. **Supplementary Table III** lists all studies excluded during full-text screening and the corresponding reasons for exclusion.

Journal Pre-proof

Patient characteristics

The 26 included studies comprised 707 patients with extracranial carotid artery aneurysms, accounting for a total of 725 carotid artery procedures. Of these, 574 were open surgical repairs (OpR) and 151 were endovascular repairs (ER). Mean patient age ranged from 46.4 to 70.2 years, and the proportion of male patients varied from 33.3% to 93.8% across studies. Baseline comorbidities, including hypertension, diabetes mellitus, dyslipidemia, coronary artery disease, smoking, and previous stroke or transient ischemic attack, were reported inconsistently among the included studies. Detailed demographic and clinical characteristics are presented in **Table I**.

Only a minority of studies provided detailed descriptions of the treatment techniques employed. When reported, endovascular treatment consisted predominantly of covered stent implantation, although bare-metal stents, flow diverters, and coil embolization were used in selected cases. Open surgical repair included aneurysmectomy with end-to-end anastomosis, aneurysmectomy with patch angioplasty, interposition grafting using autologous vein or prosthetic grafts, and carotid ligation, with interposition grafting representing the most frequently reported reconstructive technique. Technical success exceeded 90% in nearly all studies and reached 100% in several series. Antithrombotic regimens were inconsistently described across studies. Detailed treatment characteristics are presented in **Table II**.

Aneurysm characteristics, including aneurysm type, symptom status, diameter, anatomical location, bilaterality, and the presence of intraluminal thrombus, are summarized in **Table III**.

Primary comparative meta-analysis: Endovascular versus open surgical repair

Six comparative studies involving 228 procedures (55 endovascular and 173 open surgical repairs) were included in the primary meta-analysis. There was no significant difference between endovascular and open surgical repair regarding 30-day mortality (0 of 55 [0%] ER vs 1 of 173 [0.6%] OpR; OR 2.58, 95% CI 0.53–12.52; p = 0.2387; I^2 = 0%; **Figure 2**). Likewise, no significant difference was observed for 30-day stroke (OR 1.08, 95% CI 0.30–3.91; p = 0.9064; I^2 = 0%; **Figure 3).** Because five of the six comparative studies reported zero mortality events in both arms, the pooled mortality odds ratio is derived almost entirely

from a single study (Xue et al. 2020) and is highly sensitive to the continuity-correction method used for double-zero studies. Thus, the apparent discrepancy between the crude event counts (0 of 55 ER vs 1 of 173 OpR) and the primary pooled OR reflects the influence of the continuity-correction strategy in an extremely sparse-event dataset rather than a discrepancy in the extracted event counts. A sensitivity analysis using Peto's one-step odds ratio method, which does not require a continuity correction, was performed as an alternative approach for sparse-event outcomes. This analysis yielded a directionally different, non-significant estimate for 30-day mortality (Peto OR 0.21, 95% CI 0.00–13.02), in contrast with the primary continuity-corrected estimate (OR 2.58). The corresponding sensitivity analyses for 30-day stroke (Peto OR 0.45, 95% CI 0.09–2.31) and reintervention (Peto OR 1.49, 95% CI 0.22–9.98) were directionally consistent with the primary estimates. These findings indicate that the mortality odds ratio should be interpreted with caution and is not robust to the choice of rare-event meta-analytic method (Supplementary Table IV).

Observed late event rates were also similar between treatment strategies. No significant differences were identified in late stroke (OR 1.26, 95% CI 0.36–4.40; p = 0.7214; **Supplementary Figure S1),** reintervention (OR 2.19, 95% CI 0.57–8.36; p = 0.2525; **Supplementary Figure S2)**, or primary patency (OR 0.44, 95% CI 0.13–1.50; p = 0.1890; **Supplementary Figure S3)**; however, because follow-up duration was heterogeneously reported across studies and a formal time-to-event analysis could not be performed, these findings reflect similarity in cross-sectional event proportions rather than a validated comparison of long-term durability between ER and OpR. Heterogeneity was negligible across all comparative analyses ($I^2 = 0\%$).

Several prespecified outcomes — transient ischemic attack, cranial nerve injury, myocardial infarction, bleeding complications, wound and access-site complications, graft or stent occlusion, restenosis, aneurysm recurrence, secondary patency, and late mortality — could not be quantitatively pooled because of inconsistent definitions, sparse event reporting, or too few studies with extractable data. These outcomes are therefore not presented as pooled estimates and, where sufficient qualitative data were available, are discussed narratively below.

Secondary single-arm meta-analysis: Open surgical repair

Journal Pre-proof

Fourteen studies comprising 400 patients treated with open surgical repair were included in the single-arm analysis. The pooled 30-day mortality rate was 1.18% (95% CI 0.33–4.14; I^2 = 0%; **Figure 4**), corresponding to 6 deaths among 400 patients. The pooled 30-day stroke rate was 4.29% (95% CI 2.22–8.12; I^2 = 0%; **Figure 5**), based on 19 events among 400 patients.

Long-term outcomes following open surgical repair were generally favorable. The pooled late stroke rate was 0.44% (95% CI 0.03–6.20; I^2 = 0%; **Supplementary Figure S4**), corresponding to 5 events among 390 patients. The pooled reintervention rate was 2.70% (95% CI 1.35–5.48; I^2 = 0%; **Supplementary Figure S5**), based on 11 events among 399 patients. Primary patency was high, with a pooled estimate of 98.44% (95% CI 90.68–99.79; I^2 = 0%; **Supplementary Figure S6**), corresponding to 347 patent reconstructions among 363 patients. No significant heterogeneity was observed in any analysis (I^2 = 0%).

Secondary single-arm meta-analysis: Endovascular repair

Six studies including 80 patients undergoing endovascular repair were available for single-arm analysis. No 30-day mortality events were reported, resulting in a pooled event rate of 0% (95% CI 0.00–4.51%; I^2 = 0%; **Figure 6).** One 30-day stroke occurred, yielding a pooled event rate of 1.25% (95% CI 0.18–8.34; I^2 = 0%; **Figure 7**).

Long-term outcomes were similarly favorable. The pooled late stroke rate was 1.28% (95% CI 0.18–8.54; I^2 = 0%; **Supplementary Figure S7).** The pooled reintervention rate was 3.75% (95% CI 1.21–10.91; I^2 = 0%; **Supplementary Figure S8)**. Primary patency remained high at 94.50% (95% CI 84.14–99.31; I^2 = 0%; **Supplementary Figure S9).** Heterogeneity was absent across all pooled analyses (I^2 = 0%). In Xu et al., one patient who underwent internal carotid artery ligation after technical failure, without stent deployment, was excluded from the patency analysis, yielding a denominator of 14 rather than 15 patients for this outcome.

Sensitivity analyses

Leave-one-out sensitivity analyses demonstrated stable pooled estimates within the primary analytic specification for the comparative outcomes, including 30-day mortality, 30-day stroke, late stroke,

Journal Pre-proof

reintervention, and primary patency **(Supplementary Figures S10–S14)**. A sensitivity analysis using Peto's one-step odds ratio method, performed for the outcomes affected by double-zero studies (30-day mortality, 30-day stroke, and reintervention), is presented in Supplementary Figure S15 and Supplementary Table IV. Leave-one-out analyses did not materially alter the pooled estimates for most comparative outcomes. However, for 30-day mortality, the direction of the point estimate differed between the primary continuity-corrected Mantel-Haenszel analysis and the Peto sensitivity analysis, indicating substantial method dependence in the setting of a single mortality event.

Leave-one-out sensitivity analyses for open surgical repair demonstrated stable pooled estimates for both 30-day mortality and 30-day stroke, indicating that no single study exerted a disproportionate influence on the overall results **(Supplementary Figures S16–S17).**

Leave-one-out sensitivity analyses for endovascular repair demonstrated stable results for 30-day stroke, with no meaningful changes in the pooled estimates after sequential exclusion of any individual study (**Supplementary Figure S18)**. Leave-one-out analysis was not performed for endovascular 30-day mortality because no events occurred across the included studies.

Risk of bias assessment

The risk of bias assessment was performed using the ROBINS-I tool and is summarized in **Supplementary Figures S19–S21**. Overall, most included studies were judged to have a moderate to serious risk of bias. The main source of bias was confounding (Domain 1), whereas classification of interventions (Domain 2) was consistently judged to be at low risk of bias across studies. Selection of participants (Domain 3), missing data (Domain 4), and selection of the reported result (Domain 6) were predominantly judged as having moderate risk of bias, while measurement of outcomes (Domain 5) was generally considered to be at low risk.

Among the comparative studies evaluating endovascular versus open repair **(Supplementary Figure S19**), four studies (Angiletta et al., Ni et al., Pulli et al., and Szopinski et al.) were judged to have a

Journal Pre-proof

serious overall risk of bias, primarily due to confounding, whereas the studies by Choi et al. and Xue et al. were classified as having a moderate overall risk of bias.

For the single-arm analyses of open surgical repair **(Supplementary Figure S20)**, most studies were judged to have a serious overall risk of bias, largely driven by confounding and the limitations inherent to retrospective observational designs. Only the study by Illuminati et al. was classified as having a moderate overall risk of bias.

In the single-arm analyses of endovascular repair **(Supplementary Figure S21)**, four studies (Bergeron et al., Cinar et al., Obayi et al., and Welleweerd et al.) were judged to have a serious overall risk of bias, whereas Xu et al. was classified as having a moderate overall risk of bias. The study by Zaidi et al. could not be assigned an overall ROBINS-I judgment because of insufficient information. Across these studies, confounding remained the predominant source of bias, while classification of interventions and outcome measurement were generally judged to be at low risk of bias.

Publication bias was not formally assessed because fewer than 10 studies were available for most outcomes, limiting the interpretability of funnel plots and statistical tests for small-study effects.

Using the GRADE approach, the certainty of evidence for both primary outcomes, 30-day mortality (OR 2.58, 95% CI 0.53–12.52) and 30-day stroke (OR 1.08, 95% CI 0.30–3.91), was rated as very low. Both outcomes were downgraded for serious risk of bias, driven mainly by confounding by indication, and for very serious imprecision, reflecting the small comparative sample and wide confidence intervals. The full certainty assessment, including anticipated absolute effects, is presented in **Supplementary Table II.**

DISCUSSION

The rarity of ECAAs and the scarcity of randomized controlled trials make the choice between endovascular repair and open surgery particularly challenging. In this systematic review with hierarchical meta-analysis, we synthesized the available evidence by first pooling studies that directly compared the two techniques and then analyzing individual case series for each approach separately. The comparative

412 analysis included six observational studies comprising 55 endovascular procedures and 173 open surgical
413 procedures. Similar to findings reported in previous meta-analyses, we found no significant differences in
414 30-day mortality or stroke between the two strategies, and observed late event rates, including late stroke,
415 reintervention, and primary patency, were also similar between groups[4]; however, given the heterogeneous
416 follow-up reporting and the absence of formal time-to-event analyses, these results should be regarded as
417 descriptive similarity in observed outcomes rather than a validated comparison of long-term durability.
418 Statistical heterogeneity was low, and leave-one-out analyses demonstrated that no individual study
419 disproportionately influenced the pooled estimates for most outcomes, suggesting general consistency
420 across the included series; the notable exception was 30-day mortality, for which the point estimate proved
421 sensitive to the choice of rare-event meta-analytic method, as detailed below.

422 When analyzing the open surgery studies separately, we identified fourteen series involving 400
423 procedures, which demonstrated a 30-day mortality rate of only 1.18% and a perioperative stroke rate of
424 4.29%, along with a primary patency rate of approximately 98.4% (95% CI 90.7–99.8%). These findings
425 reinforce the durability of surgical reconstruction, which has long been regarded as the reference standard
426 for ECAA treatment[2,4]. According to literature reviews, aneurysm resection with restoration of blood flow
427 continues to be viewed as the approach associated with the lowest risk of late stroke[2]. Furthermore, open
428 surgery allows treatment of aneurysms with complex anatomy, particularly those located in the distal portion
429 of the artery, and enables vascular wall reconstruction using autologous or synthetic grafts when
430 necessary[2,15,16]. Although cranial nerve injury is a clinically relevant outcome when comparing treatment
431 strategies, quantitative synthesis was not possible because this endpoint was inconsistently reported
432 across studies, with heterogeneous definitions and incomplete outcome data. Available evidence was
433 therefore assessed qualitatively. Some series have reported a higher incidence of cranial nerve injury with
434 the open approach; one comparative study reported nerve injury in 14.5% of surgically treated patients,
435 whereas no such injuries occurred among patients treated endovascularly[4,5].

436 In the endovascular arm, six series totaling 80 procedures reported no cases of 30-day mortality
437 and only one 30-day stroke event, resulting in a pooled 30-day stroke rate of 1.25% with wide confidence
438 intervals. Primary patency remained high (94.5%), and reintervention rates were low (3.75%). These

findings support the short-term feasibility of endovascular repair, particularly for patients at higher surgical risk or those with aneurysms in anatomically unfavorable locations[2,3]. However, the confidence intervals around these estimates were wide because of the limited sample size and the very low number of events, indicating considerable uncertainty regarding the true event rates. Recent reviews indicate that, in larger meta-analyses, major periprocedural events occur in approximately 6% of cases, with mortality and stroke rates of 2.2% and 4.4%, respectively[6]. They also report that endovascular repair is associated with a lower stroke rate (1.1%) without reducing mortality when compared with open repair[4]. In contrast, our comparative meta-analysis did not demonstrate significant differences between treatment strategies for perioperative outcomes or in the observed proportions of late events. However, long-term follow-up data for endovascular procedures remain limited, and authors continue to emphasize the need for studies evaluating stent durability and the risk of late thrombosis or occlusion[4,6]. The excellent primary patency observed with both techniques likely reflects careful patient selection, advances in surgical reconstruction and endovascular devices, and the fact that most procedures were performed in experienced vascular centers. Nevertheless, the low number of outcome events also reduces the statistical power to detect clinically meaningful differences between treatments. Furthermore, selective publication of favorable institutional experiences cannot be excluded. To illustrate this limitation, we considered an illustrative mortality difference of approximately 1% versus 3–4%: the lower figure approximates our own pooled single-arm OpR mortality rate (1.18%), while the higher figure represents an illustrative upper-range threshold rather than a value estimated from the present comparative dataset, since the comparative analysis itself includes only a single mortality event (1 of 173 OpR patients; 0 of 55 ER patients) and cannot independently estimate a group-specific mortality rate for ER. Detecting a difference of this illustrative magnitude with adequate power would require a comparative sample of several hundred to over a thousand patients, substantially exceeding the 228 procedures available for the primary comparative analysis. Because a sensitivity analysis using Peto's method, which avoids continuity correction, reversed the direction of the point estimate (OR 0.21 versus 2.58 with the primary continuity-corrected Mantel-Haenszel approach; see Results), the numerically higher mortality point estimate observed with ER should not be read as evidence favoring either strategy; rather, it reflects the instability inherent to odds ratio estimation from very few rare events, and a substantially

larger sample would be required to determine whether either strategy carries a higher perioperative mortality risk.

The apparent equivalence between the two strategies may partly reflect selection bias inherent to observational studies. In general, endovascular repair is reserved for older patients or those at high surgical risk, with hostile cervical anatomy, a history of prior neck surgery, radiation therapy, or distally located aneurysms that are difficult to access[2,4]. In contrast, open surgery is typically performed in younger, lower-risk patients or in situations where long-term durability is prioritized[2,5]. Most included studies were retrospective and nonrandomized, and our risk-of-bias assessment identified confounding and participant selection as the main sources of bias. This confounding by indication is an inherent limitation of the available literature and likely influenced both treatment allocation and clinical outcomes.

Several limitations should be acknowledged. All data were derived from retrospective series, and cohort sizes were small, reflecting the rarity of ECAA[2,4] and increasing the risk of selection bias and residual confounding. Although statistical heterogeneity was consistently low across the pooled analyses, important clinical heterogeneity remained among the included studies, including differences in aneurysm etiology, anatomical location, treatment techniques, and follow-up duration, which may have influenced treatment selection and clinical outcomes. Outcome definitions and follow-up durations varied among studies. Although follow-up duration was reported descriptively for most series (Table I), heterogeneity and incompleteness in its reporting precluded a formal time-to-event or time-dependent analysis; our pooled estimates therefore represent cross-sectional event proportions over variable observation periods rather than validated comparisons of long-term durability between ER and OpR. Data on antithrombotic/medication regimens and on surveillance imaging protocols during follow-up were inconsistently reported or unavailable in most studies, precluding assessment of their influence on reintervention and patency. Although endovascular device type and open surgical conduit were reported for most studies (Table II), event numbers were too limited to support device- or conduit-specific subgroup analyses. In addition, the number of patients treated endovascularly was substantially smaller than the number treated with open surgery, increasing uncertainty regarding long-term estimates, particularly for endovascular repair[2,4]. Furthermore, the absence of randomized studies and the predominance of studies

judged to have moderate or serious risk of bias, mainly due to confounding, further limit the strength of the available evidence. Accordingly, when the GRADE framework was applied to the two primary outcomes, the overall certainty of evidence was rated as very low (Supplementary Table II), and these findings should therefore be interpreted with corresponding caution. Certain limitations of the review process itself should also be acknowledged. The search was restricted to three databases (PubMed, Embase, and Cochrane Library), without a dedicated search of grey literature or trial registries, and conference abstracts without sufficient data were excluded, which may have led to the omission of additional relevant studies. In addition, six potentially eligible studies published in languages other than English were excluded during full-text screening, which may have introduced language bias.

Nevertheless, the consistency of the findings, namely, the absence of significant differences in comparative outcomes, the low rates of mortality and stroke in surgical series, and the promising early results of endovascular series supports the conclusion that both techniques are associated with favorable observed perioperative outcomes and similar observed late event rates in ECAA in the available observational literature, although comparative long-term durability remains uncertain given the heterogeneity in follow-up reporting[3–5]. Open surgery remains supported by a larger evidence base and longer follow-up, whereas endovascular repair appears to be a feasible and less invasive option for selected patients in the short term, pending longer-term comparative data[2–4]. Given the predominance of retrospective studies and the substantial risk of confounding, the literature recommends the establishment of prospective multicenter registries, such as the Carotid Aneurysm Registry, to better define the natural history of the disease, refine patient selection, and more robustly compare the effectiveness and durability of different therapeutic strategies[6].

CONCLUSION

In this systematic review and meta-analysis, the available observational evidence did not demonstrate significant differences in perioperative outcomes and showed similar observed late event rates between endovascular and open surgical repair for extracranial carotid artery aneurysms, although heterogeneous follow-up and the absence of time-to-event analyses preclude a formal comparison of long-term durability. Both strategies were associated with low rates of mortality, stroke, and reintervention, as

well as excellent primary patency. Open repair remains supported by a larger body of evidence and longer follow-up, whereas endovascular repair appears to be a feasible, less invasive option with favorable short-term outcomes in appropriately selected patients, although evidence on its long-term durability remains comparatively limited. Importantly, because treatment allocation in the available studies was strongly influenced by patient characteristics, anatomical complexity, and surgical risk, the observed similarity in outcomes likely reflects confounding by indication and appropriate patient selection. Therefore, these findings should be interpreted as evidence of comparable observed outcomes in observational studies rather than proof of therapeutic equivalence between treatment strategies, and comparative long-term durability between ER and OpR remains uncertain. Further prospective multicenter studies and international registries are needed to better define patient selection, comparative effectiveness, and long-term durability.

Journal Pre-proof

REFERENCES

1. Martinez HR, Acevedo-Castillo CD, Macias-Cruz HM, Bautista-Coronado UA, Ortega-Ruiz OR, Cornejo-Hernandez A, et al. Extracranial Carotid Artery Aneurysms: A Comprehensive Analysis of its Epidemiology, Pathogenesis, Diagnosis, and Management: A Scoping Review. World Neurosurg. fevereiro de 2025;194:123528. doi:10.1016/j.wneu.2024.11.111 PubMed PMID: 39613093.

2. Hoffman ME, Squiers JJ, Hamandi M, Lanfear AT, Calligaro KD, Shutze WP. Systematic Review of the Influence of Anatomy and Aneurysm Type on Treatment Choice and Outcomes in Extracranial Carotid Artery Aneurysms. Ann Vasc Surg. julho de 2022;83:349–57. doi:10.1016/j.avsg.2022.02.006 PubMed PMID: 35257919.

3. Cornwall JW, Png CYM, Han DK, Tadros RO, Marin ML, Faries PL. Endovascular techniques in the treatment of extracranial carotid artery aneurysms. J Vasc Surg. junho de 2021;73(6):2031–5. doi:10.1016/j.jvs.2020.06.133 PubMed PMID: 33098945.

4. Galyfos G, Eleftheriou M, Theodoropoulos C, Vouros D, Georgiou K, Kimpizi D, et al. Open versus endovascular repair for extracranial carotid aneurysms. J Vasc Surg. setembro de 2021;74(3):1017-1023.e5. doi:10.1016/j.jvs.2021.04.038 PubMed PMID: 33940071.

5. Ni L, Weng H, Pu Z, Zheng Y, Liu B, Ye W, et al. Open surgery versus endovascular approach in treatment of extracranial carotid artery aneurysms. J Vasc Surg. maio de 2018;67(5):1429–37. doi:10.1016/j.jvs.2017.08.093 PubMed PMID: 29224941.

6. Willemsen SI, Jorritsma NKN, Pini R, Szeberin Z, Faggioli G, Hazenberg CEVB, et al. Endovascular and Hybrid Treatment of Extracranial Carotid Artery Aneurysms: Long Term Results from the International Carotid Aneurysm Registry. Eur J Vasc Endovasc Surg. junho de 2026;71(6):985–91. doi:10.1016/j.ejvs.2025.11.055 PubMed PMID: 41338378.

7. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ. 29 de março de 2021;372:n71. doi:10.1136/bmj.n71 PubMed PMID: 33782057; PubMed Central PMCID: PMC8005924.

8. Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al. Cochrane Handbook for Systematic Reviews of Interventions version 6.4 (updated August 2023) [Internet]. Disponível em: www.training.cochrane.org/handbook

9. U.S. National Institute of Health Research. PROSPERO. International prospective register of systematic reviews. n.d.

10. Wohlin C. Guidelines for snowballing in systematic literature studies and a replication in software engineering. Em: Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering [Internet]. New York, NY, USA: Association for Computing Machinery; 2014 [citado 16 de maio de 2024]. p. 1–10. (EASE '14). Disponível em: https://doi.org/10.1145/2601248.2601268 doi:10.1145/2601248.2601268

11. Arredondo Montero J. Fixed-Effect or Random-Effects Models? How to Choose, Perform and Interpret Meta-Analyses in Clinical Research. Journal of Evaluation in Clinical Practice. 2026;32(3):e70431. doi:10.1111/jep.70431

12. DerSimonian R, Laird N. Meta-analysis in clinical trials revisited. Contemp Clin Trials. novembro de 2015;45(Pt A):139–45. doi:10.1016/j.cct.2015.09.002 PubMed PMID: 26343745; PubMed Central PMCID: PMC4639420.

13. Methodological Standards for Meta-Analyses and Qualitative Systematic Reviews of Cardiac Prevention and Treatment Studies: A Scientific Statement From the American Heart Association. Circulation [Internet]. [citado 26 de junho de 2026]. Disponível em: https://www.ahajournals.org/doi/10.1161/CIR.0000000000000523

14. Sterne JA, Hernán MA, Reeves BC, Savović J, Berkman ND, Viswanathan M, et al. ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. BMJ. 12 de outubro de 2016;355:i4919. doi:10.1136/bmj.i4919 PubMed PMID: 27733354; PubMed Central PMCID: PMC5062054.

15. Angiletta D, Pulli R, Marinazzo D, Frotino P, Maiellaro L, Regina G. Surgical and endovascular treatment of extracranial carotid artery aneurysms: early and long-term results of a single center. Ann Vasc Surg. 2014;28(3):659–64. Located at: rayyan-594746356. doi:10.1016/j.avsg.2013.05.014

16. Szewczyk D, Andziak P, Szczerbiński J, Góra R, Mruk B, Bojakowski K. Extracranial Carotid artery aneurysm Attigah Type I Open Surgery Repair. Ann Vasc Surg. 2025;112:230–8. Located at: rayyan-594746263. doi:10.1016/j.avsg.2024.12.006

Journal Pre-proof

Table 1. Study and Patient Characteristics

First author	Year	Country	Patients (n)	Open (n)	Endovascular (n)	Age (mean)	Male (%)	Hypertension (%)	DM (%)	Dyslipidemia (%)	CAD (%)	Smoking (%)	Previous stroke/TIA (%)	Follow up
Angiletta	2014	Italy	25	23	3	55	76	NR	NR	NR	NR	NR	NR	7.5 years
Bakoyiannis	2006	Greece	9	10	0	51	77.8	NR	NR	NR	NR	NR	60	6 months–10 years
Bergeron	2004	France	4	0	5	59	75	NR	NR	NR	NR	NR	75	Mean 43 months
Choi	2022	South Korea	35	25	10	57	39.0	31.7	4.9	41.5	2.4	NR	24.4	33 months
Choudhary	2009	New Zealand	6	6	0	68.5	50	66.7	16.7	50	66.7	50	18.2	NR
Cinar	2025	Türkiye	18	0	18	46.4	44.4	NR	NR	NR	NR	NR	16.7	6–38 months
Davidovic	2004	Serbia	16	17	0	64.8	93.8	NR	NR	NR	NR	NR	58.8	Mean 5 years 3 months (range 2 months–15 years)
Faggioli	1996	Italy	20	24	0	54.8	35	30	0	10	NR	80	37.5	Mean 96.7 months
Garg	2012	USA	14	14	0	63	43	64	21	43	43	29	7.1	22 months
Illuminati	2020	Italy/France	37	37	0	58	73.0	67	27	33	33	67	13.5	16.9 years
Ni	2018	China	48	32	16	51.4	39.6	27.1	2.1	8.3	8.3	NR	16.7	46 months
Nordanstig	2014	Sweden	48	48	0	64.8	56.3	37.5	NR	NR	22.9	23.3	20.8	1 year

Journal Pre-proof

Obayi	2025	USA	27	0	29	67.6	63.0	NR	NR	NR	NR	63.0	NR	Mean 236 days
Papachristou	1996	Germany & Greece	7	9	0	69.5	42.9	71.4	14.3	14.3	42.9	57.1	55.6	1–2 years
Pulli	2013	Italy	50	47	3	70.2	74.0	82.0	22.0	46.0	30.0	50.0	26.0	60 months
Radak	2014	Serbia	84	84	0	65.0	71.4	90.5	10.7	57.1	41.7	54.8	34.5	5.0 years
Rosset	2000	France	25	25	0	54.4	88.0	NR	NR	NR	NR	NR	60.0	66 months
Skóra	2016	Poland	31	32	0	59.2	64.5	56.3	12.5	NR	25.8	62.5	57.4	Mean 68 months
Srivastava	2010	USA	19	19	0	68.3	79.0	84.0	5.0	47.0	37.0	NR	42.0	48 months
Szewczyk	2025	Poland	9	9	0	67.0	33.3	88.9	33.3	NR	22.2	22.2	22.2	41 months
Szopinski	2005	Poland	12	9	3	51.5	80.0	NR	NR	NR	NR	NR	73.3	Up to 16 years
Welleweerd	2015	Netherlands	8	0	8	52	57.1	NR	NR	NR	NR	NR	14.3	Mean 57 ± 22 months
Xu	2025	China	28	1	29	61.0	75.0	64.3	3.6	10.7	0	28.6	39.3	17.5 months
Xue	2020	China	57	37	20	52.6	43.9	70.2	17.5	NR	NR	NR	10.5	63.6 months
Zaidi	2023	USA	7	0	7	65.5	57.0	83.3	0	71.4	28.6	14.3	42.9	3.9 months
Zhang	1999	China	63	66	0	49.0	77.8	NR	NR	NR	NR	NR	20.6	6 months–15 years

NR: not reported; DM: Diabetes mellitus; CAD: coronary artery disease; TIA: transient ischemic attack

Journal Pre-proof

Table 2 Treatment characteristics of included studies

Study	Open repair (n)	Aneurysmectomy + end-to-end	Aneurysmectomy + patch	Interposition graft		Carotid ligation	Endovascular repair (n)	Covered stent	Bare-metal stent	Flow diverter	Coil embolization	Technical success (%)
				Vein graft	Prosthetic graft							
Angiletta 2014	23	15	2	5	1	1†	3	3	0	0	0	100
Bakoyiannis 2006	10	3	1	6	0	0	0	0	0	0	0	100
Bergeron 2004	0	0	0	0	0	0	5	5	0	0	0	100
Choi 2022	25	7	6	7	4	1	10	3	2	2	3	94.3
Choudhary 2009	6	2	2	1	1	0	0	0	0	0	0	100
Cinar 2025	0	0	0	0	0	0	18	0	0	18	0	94.4
Davidovic 2004	17	8	0	3	5	1	0	0	0	0	0	94.1
Faggioli 1996	24	9	5	7	3	0	0	0	0	0	0	100
Garg 2012	14	5	1	5	3	0	0	0	0	0	0	100
Illuminati 2020	37	0	0	10	27	0	0	0	0	0	0	100
Ni 2018	32	16	0	11	3	2	16	11	5	0	5	95.8
Nordansti g 2014	48	21	2	14	11	0	0	0	0	0	0	90

Journal Pre-proof

Obayi 2025	0	0	0	0	0	0	29	3	20	3	3	100
Papachristou 1996	9	1	0	3	4	1	0	0	0	0	0	88.9
Pulli 2013	47	14	9	6	18	0	3	3	0	0	0	94
Radak 2014	84	38	0	0	46	0	0	0	0	0	0	100
Rosset 2000	25	0	1	24	0	0	0	0	0	0	0	95.9
Skóra 2016	32	6	2	19	4	1	0	0	0	0	0	96.9
Srivastava 2010	19	1	6	9	3	0	0	0	0	0	0	90
Szewczyk 2025	9	7	0	1	1	0	0	0	0	0	0	100
Szopinski 2005	9	5	1	1	1	1	3	3	0	0	0	91.7
Wellewee rd 2015	0	0	0	0	0	0	8	0	7	0	1	100
Xu 2025	0	0	0	0	0	1	29	17	13	0	11	93.1
Xue 2020	37	22	0	6	4	5	20	16	4	0	0	91.5
Zaidi 2023	0	0	0	0	0	0	7	7	0	0	0	100
Zhang 1999	66	10	5	46	0	5	0	0	0	0	0	92.4

†Carotid ligation was a secondary (salvage) procedure, not the primary repair.

Journal Pre-proof

Journal Pre-proof

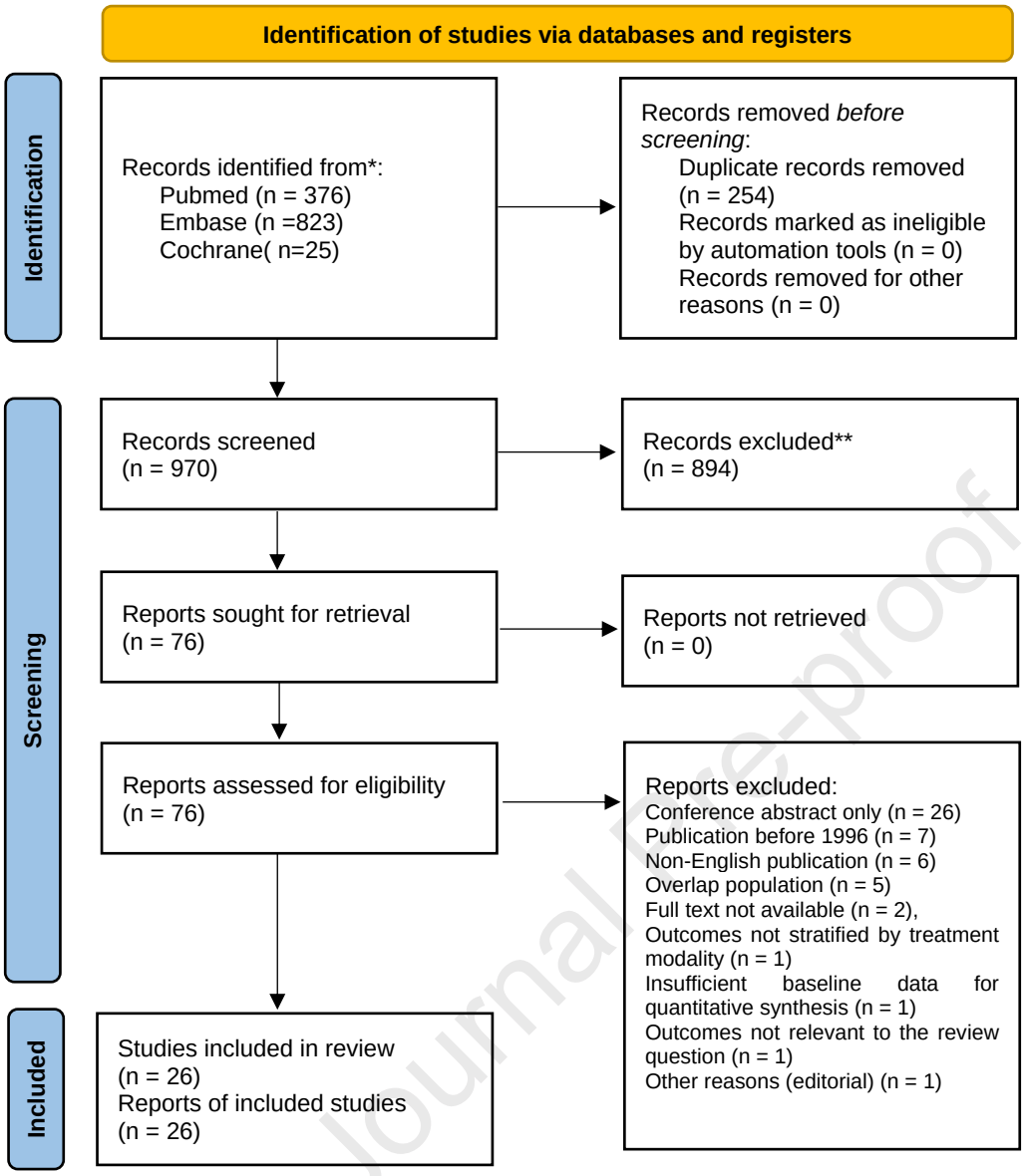
Table 3. Aneurysm Characteristics

Study	True aneurysm (n)	Symptomatic (%)	Asymptomatic (%)	Mean diameter (mm)	CCA (%)	ICA (%)	ECA (%)	Bilateral (%)	Thrombus present (%)
Angiletta 2014	18	50	50	39.7	11.5	NR	NR	4.0	NR
Bakoyiannis 2006	4	60	40	36	0	100	0	11.1	NR
Bergeron 2004	1	75	25	NR	0	100	0	25	NR
Choi 2022	24	85.9	14.1	25.0	24.4	75.6	NR	0	NR
Choudhary 2009	5	66.6	33.4	21.8	33.3	100	0	0	16.7
Cinar 2025	13	66.7	33.3	15.4	0	100	0	0	NR
Davidovic 2004	17	82.4%	17.6%	NR	11.8	88.2	0	6.3	64.7
Faggioli 1996	21	37.5	62.5	NR	20.8	100	0	9.1	NR
Garg 2012	4	7.1	92.9	24.5	42.9	35.7	0	0	NR
Illuminati 2020	37	16.2	83.8	28	0	100	0	0	100
Ni 2018	34	85.4	14.6	39.3	14.6	85.4	0	4.2	41.7
Nordanstig 2014	36	62.5	37.5	25	10.4	89.6	0	0	NR
Obayi 2025	16	31.0	69.0	22.2	27.6	58.6	0	NR	NR
Papachristou 1996	9	77.8	22.2	NR	55.6	44.4	0	28.6	NR
Pulli 2013	19	26.0	74.0	26.0	0	100	0	NR	NR
Radak 2014	80	81.0	19.0	22.6	26.2	69.1	0	17.9	72.6
Rosset 2000	21	76.0	24.0	19.1	0	100	0	NR	16.0

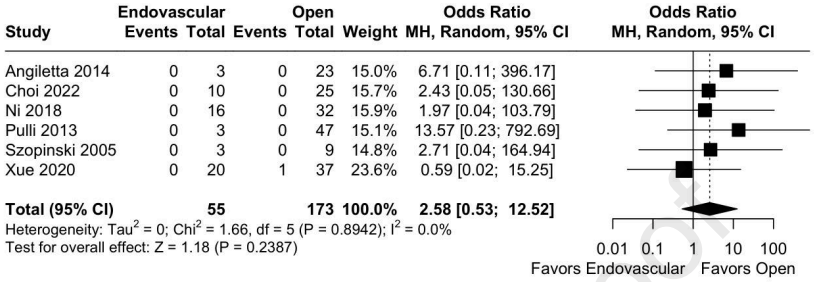
Journal Pre-proof

Skóra 2016	25	100	0	NR	56.3	43.8	NR	3.2	NR
Srivastava 2010	7	42.0	58.0	23.0	57.9	94.7	0	NR	NR
Szewczyk 2025	2	33.3	66.7	24.2	0	100	0	NR	44.4
Szopinski 2005	5	100	0	29.0	25.0	100.0	0	NR	NR
Welleweerd 2015	7	71.4	28.6	NR	0	100	0	0	0
Xu 2025	15	82.1	17.9	13.9	0	100	0	7.1	10.0
Xue 2020	45	31.6	68.4	28.0	14.0	70.2	0	0	15.8
Zaidi 2023	4	100	0	13.1	14.3	85.7	0	0	NR
Zhang 1999	37	100	0	NR	77.3	81.8	3.0	4.7	NR
CCA, common carotid artery; ECA, external carotid artery; ICA, internal carotid artery; mm, millimeters; NR, not reported.									

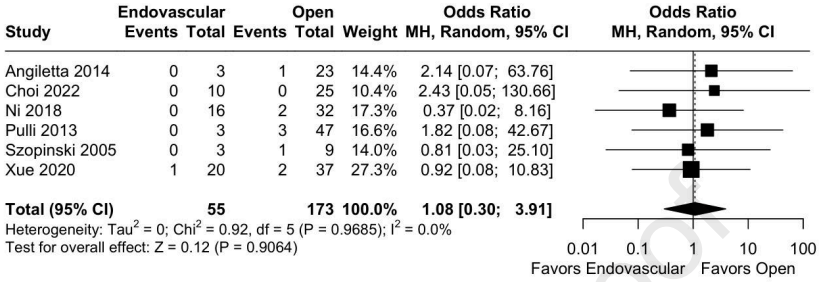
Journal Pre-proof



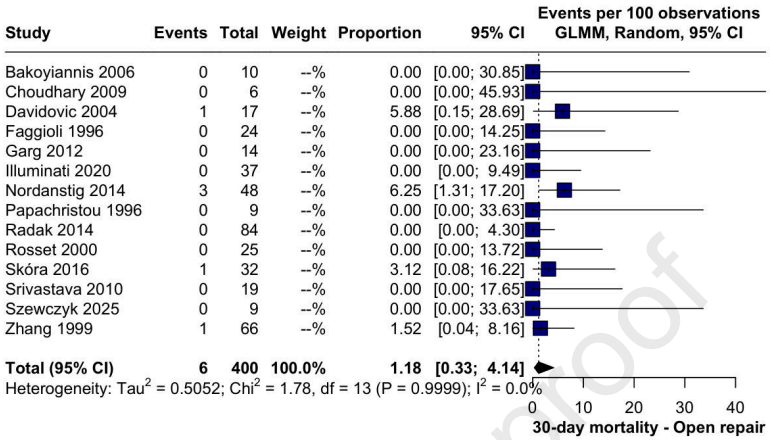
Journal Pre-proof



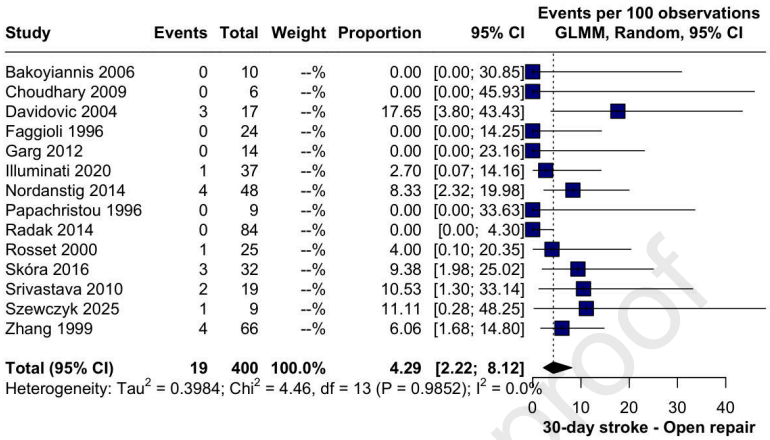
Journal Pre-proof



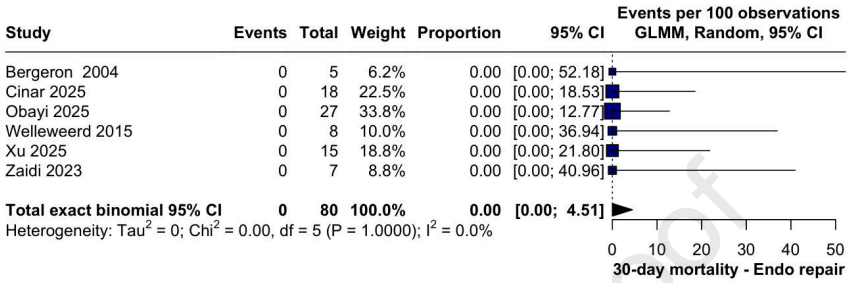
Journal Pre-proof



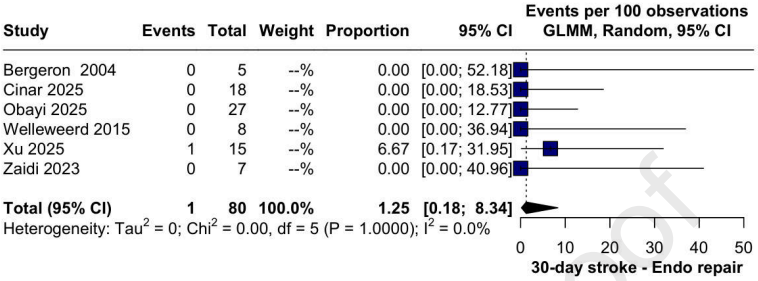
Journal Pre-proof



Journal Pre-proof



Journal Pre-proof



Journal Pre-proof

Journal Pre-proof

FIGURE LEGENDS

Figure 1. Study selection flow diagram.
PRISMA flow diagram summarizing study identification, screening, eligibility assessment, and final inclusion in the systematic review and meta-analysis.

Figure 2. Forest plot of 30d mortality comparing endovascular versus open surgical repair of extracranial carotid artery aneurysms.

Figure 3. Forest plot of 30d stroke comparing endovascular versus open surgical repair of extracranial carotid artery aneurysms.

Figure 4. Forest plot of 30d mortality following open surgical repair of extracranial carotid artery aneurysms

Figure 5. Forest plot of 30d stroke following open surgical repair of extracranial carotid artery aneurysms

Figure 6. Forest plot of 30d mortality following endovascular repair of extracranial carotid artery aneurysms.

Figure 7. Forest plot of 30d stroke following endovascular repair of extracranial carotid artery aneurysms.

Journal Pre-proof